

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 2 EXAMINATION 2023-2024****EE6402 – REAL-TIME DSP DESIGN AND APPLICATIONS**

April / May 2024

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 6 pages.
2. Answer all 5 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

1. (a)
 - (i) In IEEE 754 single-precision (32 bit) floating point format, how many bits are used for the exponent? How many bits are used for the mantissa?
 - (ii) What is the dynamic range of IEEE 754 single-precision format?
 - (iii) Determine the decimal value of the following IEEE 754 single-precision (32 bit) floating point format (normalized) number, represented by three fields—sign, exponent, and mantissa—in that order.

0	(133) ₁₀	(0.5625) ₁₀
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- (iv) Add the above floating point number to another floating point number (in the same format) shown below. Show the intermediate steps, and express the sum in the same format.

0	(131) ₁₀	(0.25) ₁₀
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(11 Marks)

Note: Question No. 1 continues on page 2.

- (b) An input $x(n)$ passes through a digital system with frequency response

$$H(e^{j\omega}) = \begin{cases} 1, & \frac{\pi}{4} \leq |\omega| \leq \frac{\pi}{2} \\ 0, & \text{otherwise} \end{cases}$$

to give an output $y(n)$. The effect of finite precision representation of $x(n)$ is modelled as a noise $u(n)$ added to the input, such that the output is now $y(n) + v(n)$, as shown in Figure 1.

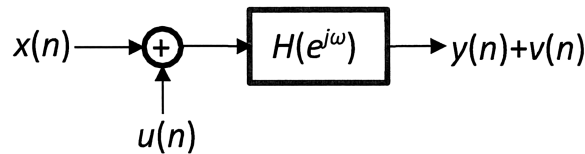


Figure 1

- (i) If $u(n)$ is the truncation error of $x(n)$, and if $u(n)$ has a variance of $\sigma_u^2 = \frac{1}{48}$, then find η_v , the mean of $v(n)$.
- (ii) If $u(n)$ is the rounding error of $x(n)$, and if $u(n)$ has a variance of $\sigma_u^2 = \frac{1}{48}$, then find σ_v^2 , the variance of $v(n)$. Also find $S_v(e^{j\omega})$, the power spectral density of $v(n)$.

(9 Marks)

2. (a) Briefly describe the difference between the mid-tread quantizer and the mid-rise quantizer. Which of these quantizers is appropriate for digitizing a speech signal that has nearly zero values for a significant number of samples, and why?

(7 Marks)

- (b)
 - (i) With the help of a diagram, explain how a successive approximation ADC (analog to digital converter) works.
 - (ii) A 3 bit successive approximation ADC is used for an input range of 0 volt to 5 volts. For an input of 1.4 volts, show the steps of how the 3 bits are determined in the ADC, and hence find the output bits.
 - (iii) What is the advantage of successive approximation ADC over counter ADC? What is the disadvantage of successive approximation ADC over flash ADC?

(13 Marks)

3. (a) Figure 2 shows a model of the sigma-delta modulator, where $x(n)$ is the input signal, $e(n)$ is the quantization error, $x_x(n)$ is the output signal, $x_e(n)$ is the output error, and the transfer functions are $P(z) = \frac{1}{1-2z^{-1}+z^{-2}}$ and $Q(z) = 2z^{-1} - z^{-2}$.

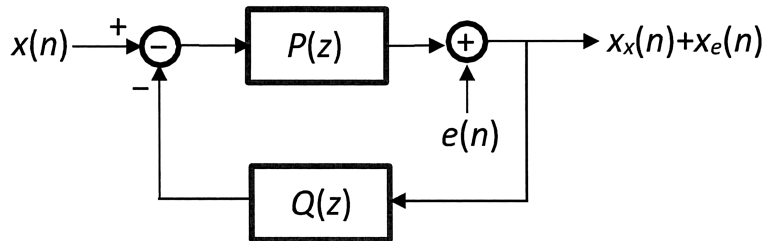


Figure 2

- (i) Neglecting the signal, show that the z -transforms of the input error and the output error are related by $X_e(z) = (1 - 2z^{-1} + z^{-2})E(z)$.
- (ii) If the quantization error $e(n)$ is a zero-mean white noise with power spectral density $S_e(e^{j\omega}) = N$, find the variance of the output error $x_e(n)$ after a (ideal) lowpass filter of passband $-B \leq \omega \leq B$. B is small enough such that in $-B \leq \omega \leq B$, $\sin \omega \approx \omega$. [Hint: Note that $1 - 2z^{-1} + z^{-2} = (1 - z^{-1})^2$. You may use $1 - \cos \theta = 2 \sin^2(\theta/2)$.]
- (10 Marks)
- (b)
- (i) A 6-bit ripple carry adder adds two 6-bit numbers using 6 full adders. If computation of each full adder requires 4 nano-seconds, find how much computation time will be required by the 6-bit ripple carry adder.
- (ii) Explain how a carry select adder reduces the computation time in part (i). Draw the block diagram of a 6-bit carry select adder by cascading 2 blocks of 3-bit adders.
- (iii) Assume the computation time for each full adder is 4 nano-seconds. How much computation time is required for each 3-bit adder block used in your carry select adder of part (ii)? Assume that each multiplexer introduces a computation delay of 2 nano-seconds. How much total computation time is required by the entire 6-bit carry select adder of part (ii)?
- (10 Marks)

4. (a) A recursive filter $y(n) = x(n) - 0.9y(n-1]$ is implemented as shown in Figure 3(a). The computation times of the adder and the multiplier are shown in brackets.

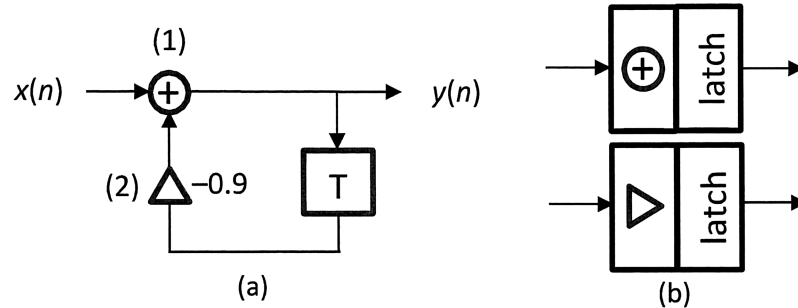


Figure 3

- (i) The implementation of Figure 3(a) uses 1 adder, 1 multiplier, and has latency of 1. What is the critical path of this implementation?
 - (ii) In order to reduce the critical path, use pipelining such that, a latch is inserted in each adder, and a latch is inserted in each multiplier, as shown in Figure 3(b). Rewrite $y(n)$ appropriately, and draw the pipelined implementation. Show all multiplier values in your implementation. Let the output of your pipelined implementation be $y'(n)$. Express $y'(n)$ using $y(n)$, that is, find m where $y'(n) = y(n-m)$.
 - (iii) Compare the number of adders, the number of multipliers, latency, and critical path of your pipelined implementation in part (ii) to the implementation in part (i).
- (12 Marks)
- (b)
- (i) Describe the polyphase parallel FIR filter implementation for 2 polyphase components. Let the transfer function of the FIR filter be $H(z) = 0.1 + 0.2z^{-1} + 0.3z^{-2} + 0.3z^{-3} + 0.2z^{-4} + 0.1z^{-5}$. Find its 2-phase polyphase components.
 - (ii) Compare the number of 'multiply' required by the conventional implementation of $H(z)$, to the number of 'multiply' required by your polyphase implementation in part (i). If the polyphase parallel FIR filter implementation increases the required number of 'multiply' (and 'add'), what is the advantage of the polyphase parallel FIR filter implementation?
- (8 Marks)

5. (a) An algorithm has to be implemented on a DSP processor to process a block of n_x input samples. A part of the algorithm requires FIR filtering on the input. A maximum of 1000 CPU cycles may be used for the FIR filtering. The DSP processor requires $n_x(3 + (n_h/2)) + 25$ CPU cycles to compute n_x output samples of an FIR filter, where n_h is the number of filter coefficients.
- Explain a possible reason why the required number of CPU cycles contains $n_h/2$ and not n_h . Also explain possible reasons why 3 CPU cycles per input sample, and 25 additional CPU cycles are required.
 - If a block has $n_x = 25$ input samples, then what is the maximum number of filter coefficients n_h that can be accommodated within 1000 CPU cycles?
 - If n_x may take any value from 13 to 32, then what is the maximum number of filter coefficients n_h , and the corresponding number of input samples n_x , that can be accommodated within 1000 CPU cycles?
- (11 Marks)

- (b) In TMS320C55x, a circular buffer of size 8 has been defined in memory locations 01_0160 to 01_0167. The circular buffer and the data in it are shown below.

memory location	data
01_0160	0005h
01_0161	0007h
01_0162	0001h
01_0163	0006h
01_0164	0000h
01_0165	0004h
01_0166	0002h
01_0167	0003h

Auxiliary register AR0 is used as the pointer to this circular buffer. Initial values of AR0 and two temporary registers are shown below.

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XAR0 01_0160h
T0    0000h
T1    0000h

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Note: Question No. 5 continues on page 6.

Using circular buffer addressing rule, find the values of the mentioned registers after each of the following consecutive instructions:

MOV *AR0, T0	find the value of T0
MOV *(AR0+T0), T1	find the values of XAR0 and T1
MOV *(AR0+T1), T0	find the values of XAR0 and T0
MOV *AR0(T0), T1	find the values of XAR0 and T1
MOV *AR0(T1), T0	find the values of XAR0 and T0

(9 Marks)

END OF PAPER

EE6402 REAL-TIME DSP DESIGN & APPLICATIONS

Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.